Alibaba Cloud

We run an online store in Southeast Asia with about 30,000 daily users and big sales peaks

Architecture proposal

Date	Revisions	Products	Catalog checked
2 November 2026	1	11	2026-10-31

Prepared with Architecture Builder

AI-generated. Review with a solutions architect before you build.

Summary

The design

I'll design a highly available two-zone web store in Singapore, fronted by ESA and ALB with WAF, ECS behind Auto Scaling for sales peaks, Tair cache and RDS. Sessions go to Tair and orders to RDS, both deployed across two zones for resilience.

11 products, 8 connections, 1 revisions

Pinned requirements

No requirements were pinned.

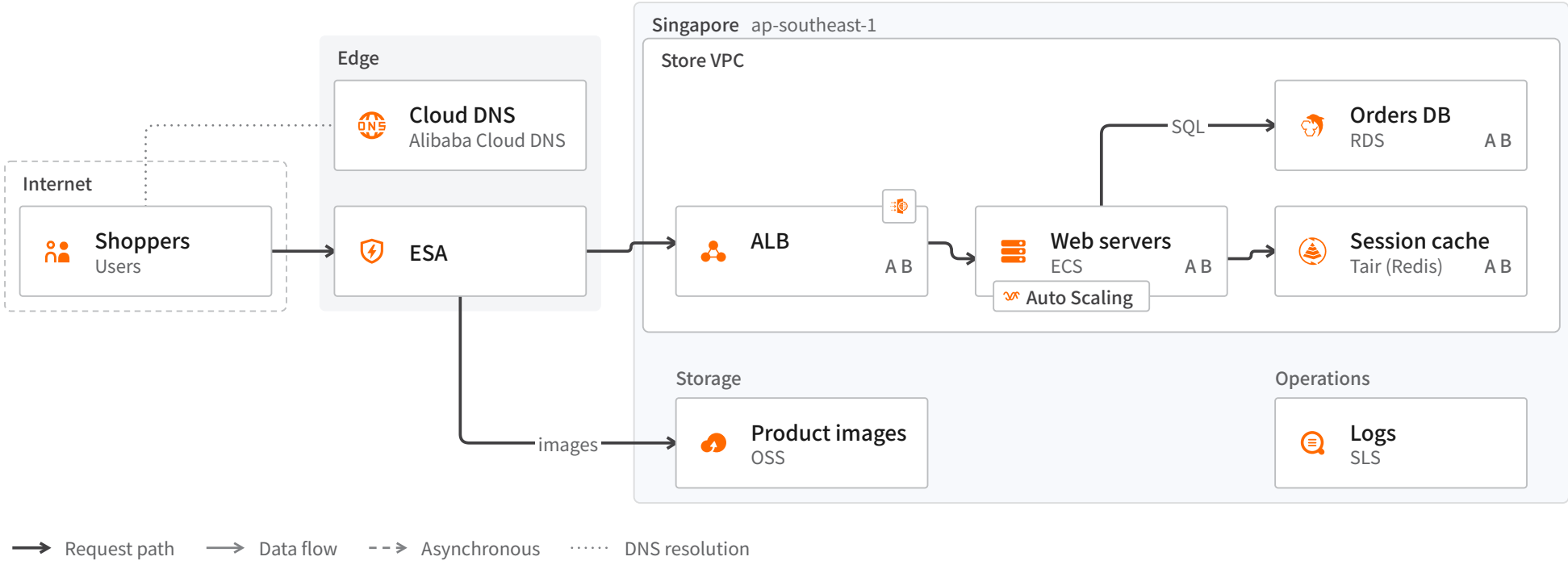
Constraints

No region or product constraints were set.

Assumptions

No assumptions were recorded.

Architecture diagram



Components

Every product in the diagram, why it is in this design, and where to read its official documentation.

Users

Shoppers
Internet

Role in this design
30,000 daily users browsing the online store.

What it does
People reaching the system over the internet from browsers or apps.

Official documentation
Documentation link not yet available

Alibaba Cloud DNS

Edge · Domain Names and Websites

Role in this design
Resolves the store domain to ESA edge nodes.

What it does
Authoritative public DNS for your domains; a resolve hop, not on the data path.

Official documentation
www.alibabacloud.com/product/dns

Edge Security Acceleration (ESA)

Edge · Networking & CDN

Role in this design
Caches static assets and accelerates dynamic pages across Southeast Asia.

What it does
Global edge: caching, dynamic acceleration, WAF, DDoS defense, bot management and edge functions.

Official documentation
www.alibabacloud.com/product/esa

Application Load Balancer (ALB)

Singapore (ap-southeast-1) · Store VPC · Zones A, B

Role in this design
Distributes traffic across two zones for high availability.

What it does
Layer-7 HTTP/HTTPS/gRPC load balancer with content routing; also serves as ACK Ingress.

Official documentation
www.alibabacloud.com/product/server-load-balancer

Web Application Firewall (WAF)

Singapore (ap-southeast-1) · Store VPC · On ALB

Role in this design
Blocks OWASP attacks and bots before reaching the app.

What it does
WAF 3.0: blocks OWASP attacks, CC floods and bots via cloud-native or CNAME access.

Official documentation
www.alibabacloud.com/product/web-application-firewall

Elastic Compute Service (ECS)

Web servers
Singapore (ap-southeast-1) · Store VPC · Zones A, B

Role in this design
Runs the store application across two zones.

What it does
Virtual machines (x86, Arm, GPU) with cloud disks and security groups.

Official documentation
www.alibabacloud.com/product/ecs

Auto Scaling (ESS)

Singapore (ap-southeast-1) · Store VPC · On Web servers

Role in this design
Adds instances automatically during big sales peaks.

What it does
Scales ECS or ECI fleets by rules, schedules or metrics and registers them to load balancers.

Official documentation
www.alibabacloud.com/product/auto-scaling

ApsaraDB RDS

Orders DB
Singapore (ap-southeast-1) · Store VPC · Zones A, B

Role in this design
Stores orders with a standby replica in the second zone.

What it does
Managed MySQL, PostgreSQL, SQL Server and MariaDB with HA, read replicas and backups.

Official documentation
www.alibabacloud.com/product/apsaradb-rds

Tair (Redis® OSS-Compatible)

Session cache
Singapore (ap-southeast-1) · Store VPC · Zones A, B

Role in this design
Keeps carts and sessions fast with zone-level HA.

What it does
Managed Redis-compatible in-memory database and cache, plus Tair enterprise features.

Official documentation
www.alibabacloud.com/product/tair

Object Storage Service (OSS)

Product images
Singapore (ap-southeast-1) · regional services · Storage

Role in this design
Stores product images that ESA serves from the edge.

What it does
Durable object storage for files, media, backups and data lakes; S3-compatible.

Official documentation
www.alibabacloud.com/product/object-storage-service

Simple Log Service (SLS)

Logs

Singapore (ap-southeast-1) · regional services · Storage

Role in this design

Centralizes logs and alerts for monitoring store health.

What it does

Collects, searches, analyzes with SQL and alerts on logs, metrics and traces.

Official documentation

www.alibabacloud.com/product/log-service

Connections

How the products in the diagram talk to each other.

From	To	Carries	Type
Shoppers	Cloud DNS		DNS resolution
Shoppers	ESA	HTTPS	Data
ESA	ALB	origin	Data
ALB	Web servers		Data
Web servers	Session cache		Data
Web servers	Orders DB	SQL	Data
ESA	Product images	images	Data
Web servers	Logs		Control

Revision history

Every ask and what it changed in the design.

Revision 1

Drawn in 17.4 s

Ask

We run an online store in Southeast Asia with about 30,000 daily users and big sales peaks. Design a highly available web architecture.

Changes

+ Users · Shoppers · in Internet
+ Cloud DNS
+ ESA · in Edge
+ ALB · zones A, B
+ WAF · on ALB
+ ECS · Web servers · zones A, B
+ Auto Scaling · on ECS
+ Tair (Redis) · Session cache · zones A, B
+ RDS · Orders DB · zones A, B
+ OSS · Product images · in Singapore (ap-southeast-1)
+ SLS · Logs · in Singapore (ap-southeast-1)
+ 8 connections